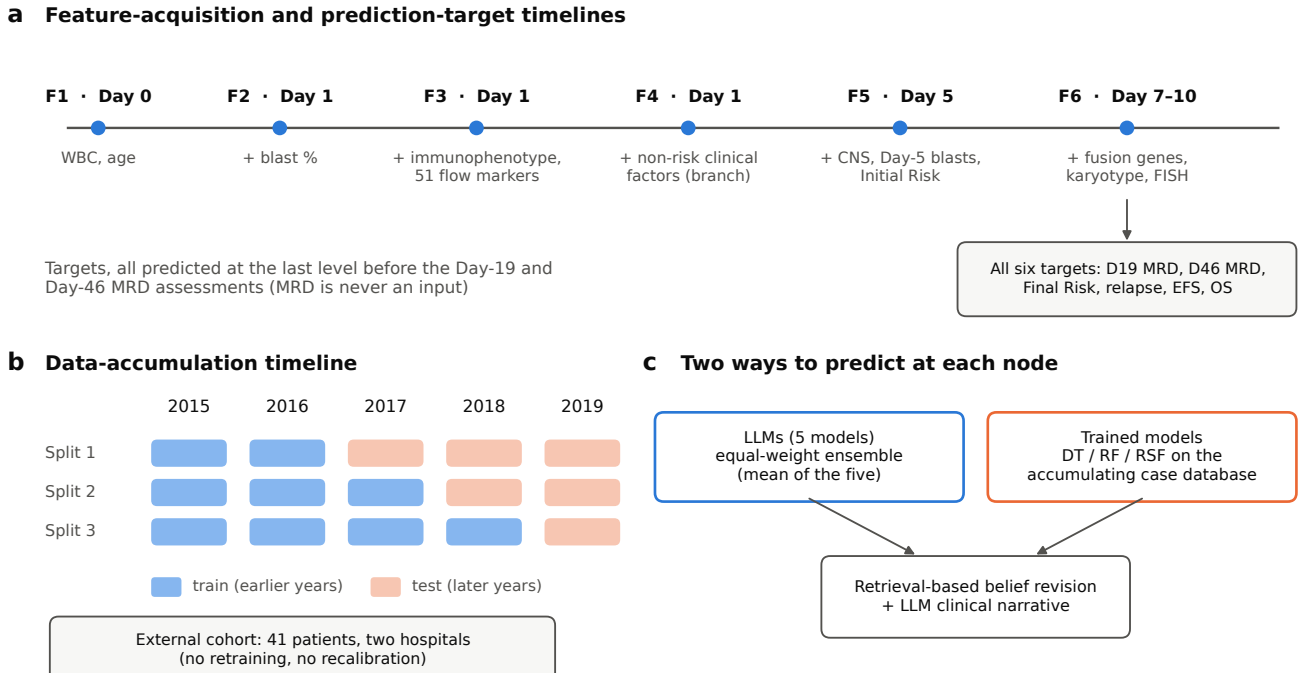
a Feature-acquisition and prediction-target timelines
F1 · Day 0
WBC, age
F2 · Day 1
+ blast %
F3 · Day 1
+ immunophenotype, 51 flow markers
F4 · Day 1
+ non-risk clinical factors (branch)
F5 · Day 5
+ CNS, Day-5 blasts, Initial Risk
F6 · Day 7–10
+ fusion genes, karyotype, FISH
Targets, all predicted at the last level before the Day-19 and Day-46 MRD assessments (MRD is never an input)
All six targets: D19 MRD, D46 MRD, Final Risk, relapse, EFS, OS
b Data-accumulation timeline
2015
2016
2017
2018
2019
Split 1
Split 2
Split 3
train (earlier years)
test (later years)
External cohort: 41 patients, two hospitals (no retraining, no recalibration)
c Two ways to predict at each node
LLMs (5 models) equal-weight ensemble (mean of the five)
Trained models DT / RF / RSF on the accumulating case database
Retrieval-based belief revision + LLM clinical narrative